

we find a negative and statistically significant difference in differences estimate for the complementary group for novice workers. This indicates that the demand for novice labor decreased in the complemented cluster after the launch of ChatGPT. This finding highlights the importance of considering other dimensions of heterogeneity beyond skill requirements when studying the effects of LLMs on the labor market.

3.2.3. Effect heterogeneity by skill cluster

Next, we formally examine the heterogeneity of the impact of the ChatGPT launch across clusters in the substitutable and complementary AI exposure categories by estimating the regression in Eq. (2) presented in the methods section.

Fig. 5 illustrates the estimated changes in demand for each skill cluster in the substitutable and complementary category. Most substitutable clusters exhibit a decline compared to the counterfactual: Writing related to real estate and 'about us' pages show the most substantial decreases, at 52% and 59%, respectively. The largest cluster in our sample of affected skill clusters—content writing for blogs—declined by 20%. Translation is also significantly impacted, with the 'Western European Languages' category experiencing a 23% decline. In the complementary clusters, demand changes are mixed, with some clusters showing significant decreases and others increases. Notably, the demand for jobs in the cluster 'AI-powered chatbots' nearly tripled, with a 179% increase. The cluster of 'Machine learning' also shows a significant rise, with demand increasing by 29%.

Cluster-specific estimates should be interpreted with caution for two reasons. First, while the complementary and substitutable clusters show no statistically significant pre-trends on average, individual clusters may still display notable pre-trends, which could influence some results. Rather than excluding clusters with statistically significant pre-trends, we present results for all clusters. Given the large number of comparisons, some statistically significant pre-trends will occur by chance, and it is difficult to distinguish between true pre-trends and those arising randomly. Second, comparing estimates requires adjusting for multiple hypothesis testing. However, adjusting for every pairwise comparison is impractical, as it would inflate standard errors and make Fig. 5 uninformative.⁴ We therefore recommend focusing on the top and bottom of the distribution shown in Fig. 5.

Notwithstanding these caveats, the main takeaway from Fig. 5 is that ChatGPT resulted in substantial demand increases for clusters related to AI for chatbot development, lead generation, and machine learning and significantly decreased demand for writing and translation jobs.⁵

The results from Fig. 5 align with results on the performance of LLMs across languages (Li et al., 2024) and domains (del Rio-Chanona et al., 2024). The significant decline in demand for translation of Western European languages compared to less common languages, such as Arabic and Hebrew, is consistent with the superior performance of LLMs in Western languages, making them more easily substitutable. Clusters related to writing that do not exhibit a significant decrease in demand include Spanish localization and translation, fiction writing, editing, and script writing for explainer videos. These skills require an understanding of language nuances, creativity, and interaction with video content, which explains their relatively stable demand. In programming, the most affected clusters are related to JavaScript, HTML, and CSS, which are also among the most affected programming languages in collaborative programming platforms such as Stack Overflow (del Rio-Chanona et al., 2024). The increase in demand for AI-powered chatbot development and machine learning is intuitive, as businesses increasingly adopt these technologies.

3.2.4. Sensitivity analyses

We have conducted a series of sensitivity analyses to demonstrate that our results are robust to reasonable variations in the regression model and pass standard checks for identification. While the main exhibits related to these exercises are provided in the appendices, we briefly discuss each of them here individually.

Placebo tests. We first provide a standard 'placebo treatment time' test to show that our results do not capture underlying trends that are unrelated to ChatGPT. We do this by entirely dropping the post-ChatGPT period from our data. Thereafter, we introduce a placebo treatment set to 30th May 2022. In Appendix F, we show that the difference in differences estimates are indistinguishable from zero in this case.

Dropping graphic design. So far, we have focused on studying the effects of LLMs on the labor market, ignoring other breakthroughs in generative artificial intelligence technologies related to image generation (such as Dall-E, originally released in 2021, and Midjourney, originally released in 2022). According to our labeling which only focuses on text-based models, graphic design tasks are mostly in unaffected clusters. Nonetheless, one could argue that our results might be affected by graphic design jobs in the control group, which were likely affected by the advent of advanced AI image generation capabilities. We addressed this by dropping all clusters that include 'graphic design' as one of their important skills from the control group. We report the results of this robustness check in Appendix G, which are virtually unchanged.

⁴ Depending on the exact method for multiple hypothesis testing correction, we would need to inflate the standard errors by some constant for every 70×69 pairwise comparison we make.

⁵ The standard errors on cluster-specific estimates are very similar because the analysis sample consists of week-by-cluster cells, resulting in identical sample sizes across clusters.